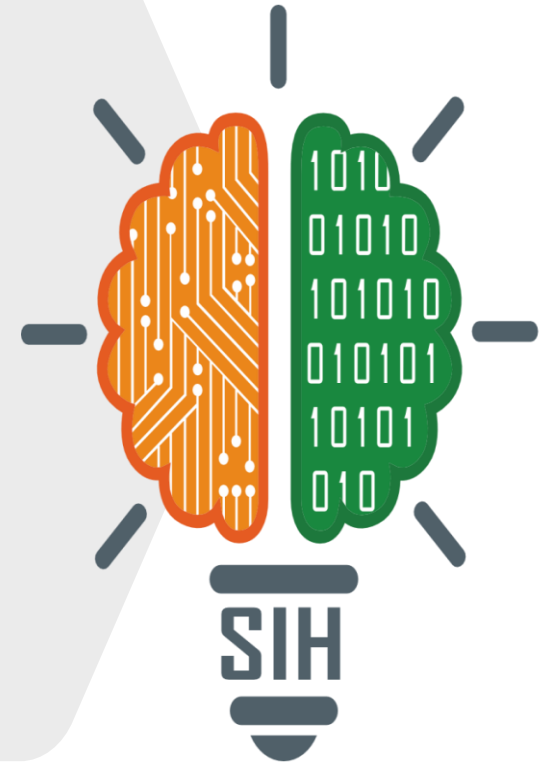


SMART INDIA HACKATHON 2026



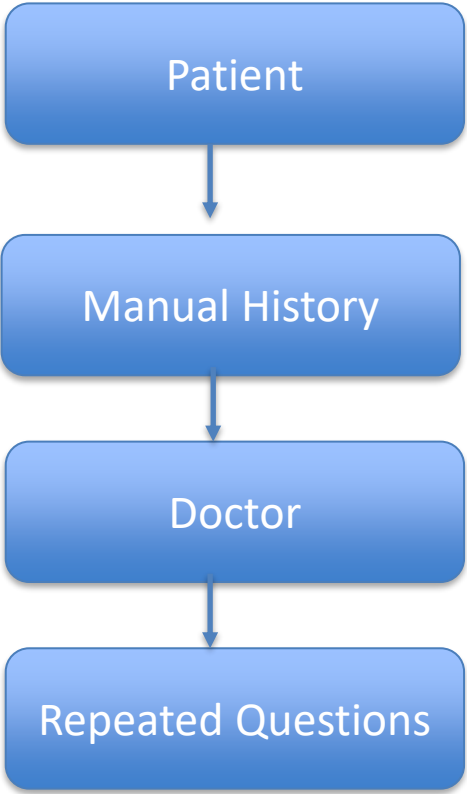
TITLE PAGE

- **Problem Statement ID –26047**
- **Problem Statement Title-PATIENT CASE-TAKING SOFTWARE**
- **Theme-Healthcare / MedTech**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal):Quantum Coders**

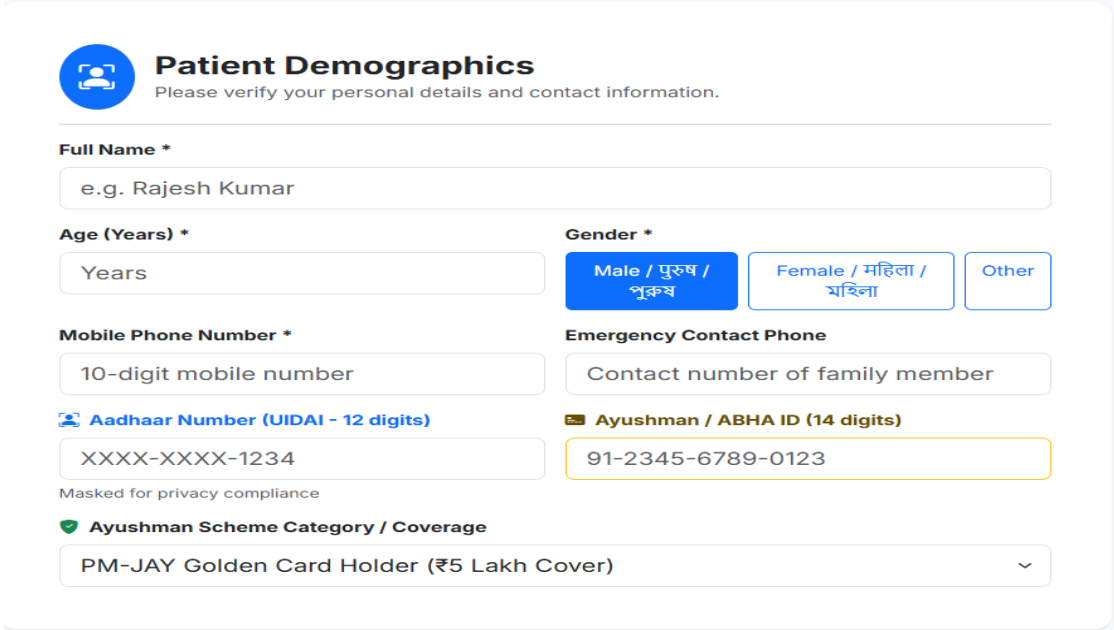
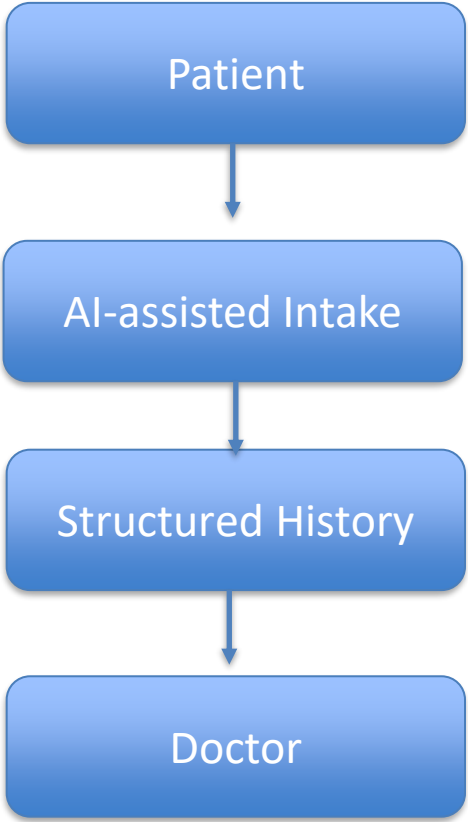


IDEA TITLE

CURRENT BOTTLENECK



MEDIKIOSK



Patient Demographics
Please verify your personal details and contact information.

Full Name *
e.g. Rajesh Kumar

Age (Years) *
Years

Gender *

Mobile Phone Number *
10-digit mobile number

Emergency Contact Phone
Contact number of family member

 Aadhaar Number (UIDAI - 12 digits)
XXXX-XXXX-1234

 Ayushman / ABHA ID (14 digits)
91-2345-6789-0123

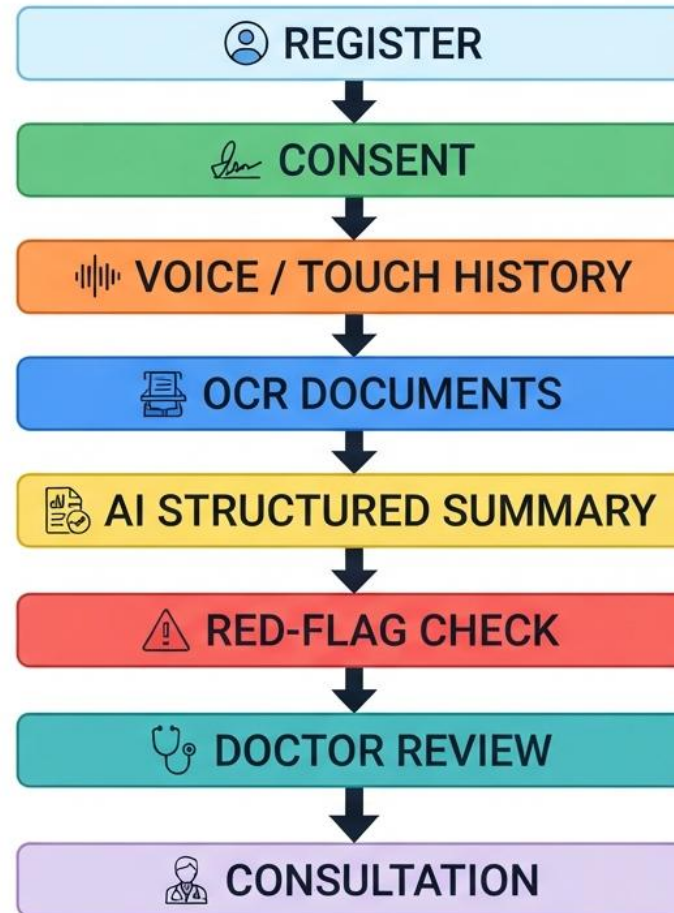
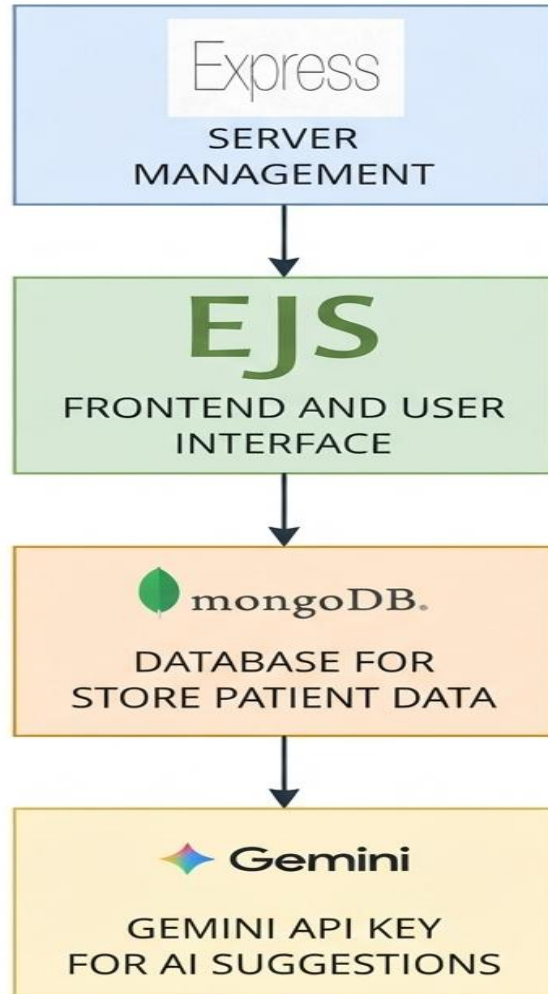
Masked for privacy compliance

 Ayushman Scheme Category / Coverage
PM-JAY Golden Card Holder (₹5 Lakh Cover)

- 1.Reduces repetitive history-taking workload
- 2.Digitizes previous medical documents using OCR
- 3.Provides a structured clinical summary before consultation

Quantum
Coders

TECHNICAL APPROACH



Active Patient Queue

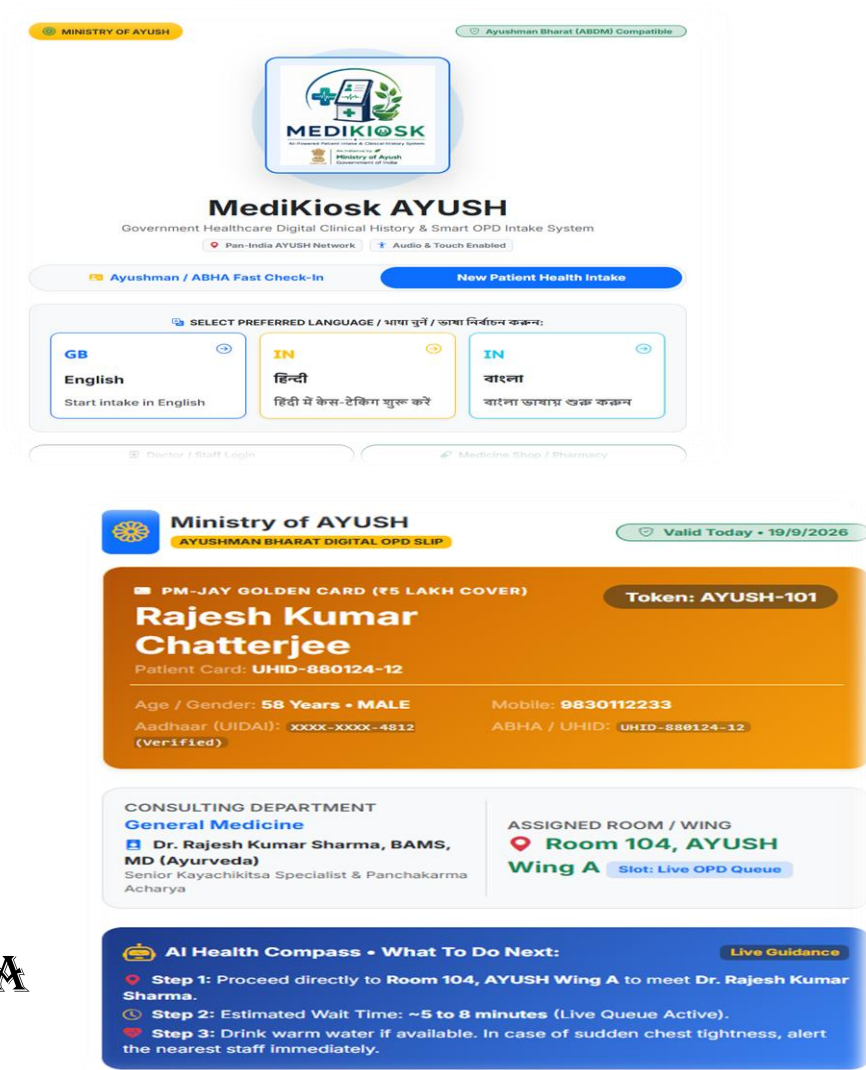
Filter patient or token...

PRIORITY	TOKEN	PATIENT NAME	AGE / GENDER	PHONE	STATUS	CONSULTATION ACTION
Red-Flag	AYUSH-113	Ayush Patient UHID: UHID-AYUSH-123845-712 • Lang: BN	35 yrs / other	9999999999	Ready for Review	Open Case File
Red-Flag	AYUSH-110	Walk-In Checkin Patient UHID: UHID-AYUSH-357178-786 • Lang: EN	29 yrs / female	9876001122	Ready for Review	Open Case File
Red-Flag	AYUSH-117	Rajesh Kumar Chatterjee UHID: UHID-880124-12 • Lang: BN	58 yrs / male	9830112233	Ready for Review	Open Case File

OUR APPLICATION SEAMLESSLY HANDLES MULTIPLE PATIENT REQUESTS SIMULTANEOUSLY, ENSURING A SMOOTH AND USER-FRIENDLY EXPERIENCE FOR EVERYONE.

Challenge	Risk	Our Strategy
Noisy hospital environment	Voice recognition errors	Voice + touch fallback
AI hallucination	Incorrect clinical information	Doctor verification
Low digital literacy	Patient unable to use kiosk	Guided UI + audio + multilingual flow
Medical document quality	OCR errors	OCR treated as draft + human verification
Sensitive health data	Privacy/security risk	RBAC + authentication + audit logs

ONCE THE PATIENT COMPLETES THE SYSTEM REGISTRATION, A SECURE CARD IS GENERATED TO HOLD ALL OF THEIR CORE MEDICAL AND PERSONAL DATA.



Quantum
Coders

IMPACT AND BENEFITS



Better information before consultation

Structured history
Previous reports in one place
Less repetitive questioning



Simpler & more accessible intake

Voice interaction
Touch-based input
Multilingual workflow
Better continuity for returning patients



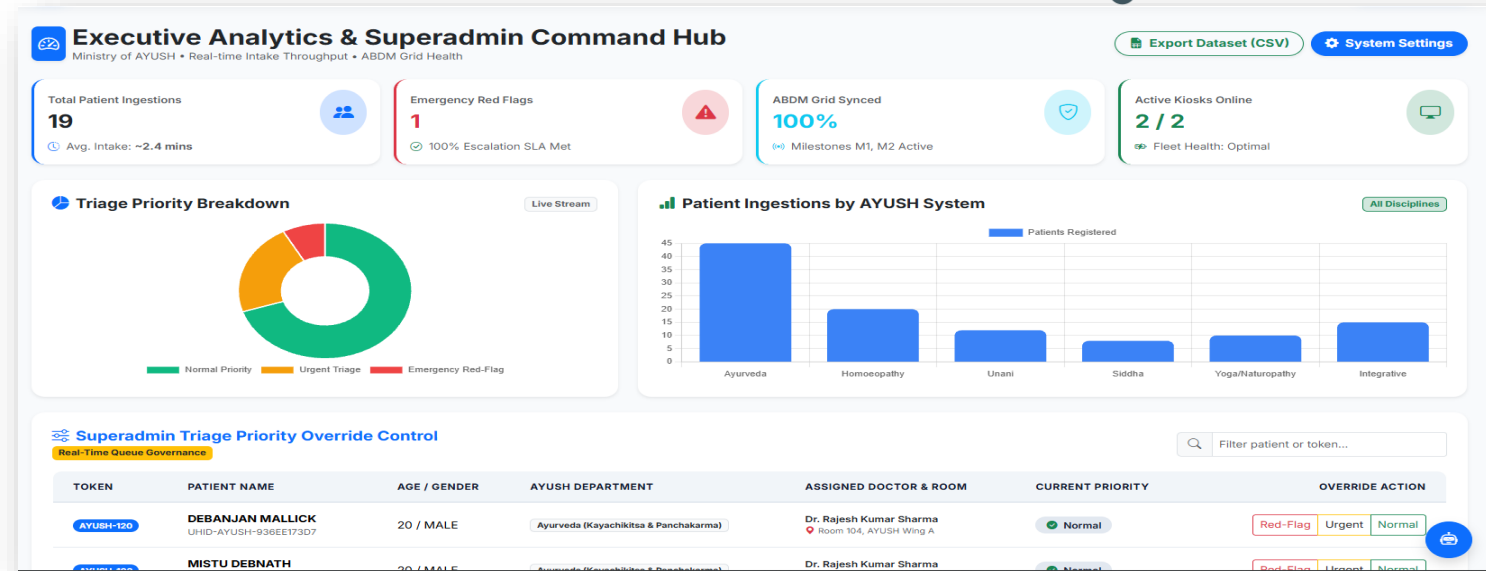
Connected workflow

Digital patient records
Queue management
Pharmacy workflow
Reports & documents
Role-based access



Scalable digital foundation

Standardized clinical intake
Reduced manual data entry
Potential HIS/ABDM integration
Better digital continuity



Doctor Consultation Queue
Department: General Medicine • Clinician: Dr. Arindam Banerjee (MD, Internal Medicine)

14 Patients in Queue • Add Walk-in

Active Patient Queue

PRIORITY	TOKEN	PATIENT NAME	AGE / GENDER	PHONE	STATUS	CONSULTATION ACTION
Red-Flag	AYUSH-115	Ayush Patient UHD: AYUSH-02345-789 • Lang: EN	35 yrs / other	9999999999	Ready for Review	Open Case File
Red-Flag	AYUSH-116	Walk-in Check-in Patient UHD: AYUSH-20789-789 • Lang: EN	29 yrs / female	9876543210	Ready for Review	Open Case File
Red-Flag	AYUSH-117	Rajesh Kumar Chatterjee UHD: AYUSH-01234-567 • Lang: EN	58 yrs / male	983072233	Ready for Review	Open Case File
Normal	AYUSH-120	DEBANJAN MALLICK UHD: AYUSH-936E173D7 • Lang: EN	20 yrs / male	878878905	Ready for Review	Open Case File
Normal	AYUSH-103	MISTU DEBNATH UHD: AYUSH-08070405E • Lang: EN	20 yrs / male	7602843724	Ready for Review	Open Case File
Normal	AYUSH-119	MISTU DEBNATH UHD: AYUSH-03880-487 • Lang: EN	20 yrs / male	7602843723	Ready for Review	Open Case File

Showing Top 6 of 14 entries

AYUSH Medicine Shop & Dispensary Hub
Prescription Retrieval • All Drug Allergy Safety Shield • Point-of-Sale (POSD) Dispensation

Fetch Doctor's Prescription by Token / UHD / Mobile

Enter Token (e.g. AYUSH-101), UHD, or Patient Mobile

Retrieve Rx

Recent Patients: AYUSH-120 (DEBANJAN), AYUSH-103 (MISTU), AYUSH-117 (MISTU), AYUSH-116 (Ayush), AYUSH-115 (Ayush)

Inventory Control
Add Stock Item

MEDICINE	BATCH	STOCK	EXPIRY	ACTION
PARACETAMOL Medicine	8555555	25	Not set	Adjust

Receive Stock Delivery
Receiver is recorded automatically

Medicine name * Batch number * Quantity * Supplier *

Invoice number * Truck / vehicle number * Delivery person * dd-mm-yyyy *

Unit price * Receiving notes *

Receive

MORE TIME FOR CLINICAL REASONING. LESS TIME COLLECTING INFORMATION.⁵

Clinical Consultation Time

Irving et al., BMJ Open (2017)

International variations in primary care physician consultation time: a systematic review of 67 countries

- Reviewed **179 studies / 111 publications**
- Covered approximately **28.5 million consultations**
- Reported substantial international variation in consultation duration.
- The study found that **18 countries representing about 50% of the global population had reported consultation times of 5 minutes or less.**
- Supports the need to reduce avoidable information-collection workload during consultations.

[BMJ Open Research Paper](#)

WHO Guideline — Recommendations on Digital Interventions for Health System Strengthening

WHO evaluates digital interventions using factors such as:

- Benefits and potential harms
- Acceptability
- Feasibility
- Resource requirements
- Equity
- Implementation considerations

WHO also emphasizes that digital interventions should **support rather than substitute functioning health systems.**

[WHO Digital Health Guideline](#)